INCH-POUND
MIL-DTL-24643/79
27 November 2023

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +221 DEGREES FAHRENHEIT (+105 DEGREES CELSIUS), 2,000 VOLTS, 133 PERCENT RATED, TYPE LS2KVTSGSG

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Overall Shielded with Circuit Integrity)

Component 1 (three unshielded conductors):

First - Copper conductor, 127/0.057 inches uncoated, 0.752 inch nominal.

Second - Uniformly extruded silicone-glass tape insulation, 0.984 inch nominal thickness, with an extruded covering of polyester elastomer – white (natural), 1.008 inches nominal wall thickness.

Component 2 (three insulated ground wires):

Third - Copper conductor, coated or uncoated, 19/0.0372 inches tin copper, 0.184 inch nominal wall thickness.

Fourth - Uniformly extruded thermoset EPR (class IV, white) insulation, 0.244 inch nominal wall thickness.

Assembly:

Fifth - The three component 1 unshielded conductors and three component 2 ground wires shall be cabled together with a lay not greater than 24 times the pitch diameter. Cable core shall be filled with soft filler to form a firm, well-rounded assembly and wrapped with a mylar binder tape spirally applied with overlap sequential marker tape pulled in under binder.

Sixth - 32 AWG braided tin copper shield. 85 percent minimum coverage.

Seventh - Mylar binder tape spirally applied with overlap.

Eighth - Cross-linked polyolefin jacket (black). Absolute minimum average wall thickness 0.075 inch. See table I for cable overall diameter.

Aramid yarn rip cords pulled in under jacket
Cable Outer Diameter: 2.340 inches, nominal
Weight: 5.60 lb/ft, nominal
Bend Radius: 28.75 inches, minimum

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/79	Type and size	Conductor size/ ground sizes		Cable overall diameter max. (inch)	Conductor resistance (ohms)	Insulation resistance (megohms)	Accelerated service loading (amperes)
		Navy STD	AWG / MCM				
-01UO	LS2KVTSGSG-400	400 (127)	400 MCM	2.34	0.0273/ Insulated ground wires (Gnds): 0.449	250	750

REQUIREMENTS:

Qualification required.

INSPECTION:

Bending radius (permanent training) (min.)	Shall be 8 times the cable maximum specified overall diameter. The term “minimum bend radius” indicates the smallest diameter circle around which the cable may be bent, resulting in the maximum amount of cable bend allowed. This bend radius applies to cable being bent in place, with no tension applied to the cable or conductors.
Basic electricals:	
Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	See table I
Voltage withstand (volts, root mean square [Vrms], min.)	
Conductor to conductor	9,500
Conductor to ground wire	9,500
Conductor to shield	9,500
Ground wire to shield	2,000
Insulation resistance (megohms/1,000 feet, min.)	See table I
Conductor to conductor	225
Conductor to ground wire	225
Conductor to shield	225
Ground wire to shield	See table I
Conductor, ground, and shield continuity	No failure
Jacket flaws	No failure

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Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation (extruded insulation only)	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure
Group D:	
Gas flame (see NOTE below) (circuit integrity) (1 hour)	No failure
Flame propagation (cable)	No failure

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Qualification inspection:

Qualification inspection shall include basic electrical; all of groups A, B, C, and D; plus the following:

Cold working (-4±3.6 °F [-20±2 °C])	No damage
Gas flame (see NOTE below) (circuit integrity) (3 hours)	No failure
Accelerated service (see table I for load current)	No failure
Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Semi-conducting materials	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Semi-conducting materials	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Semi-conducting materials	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Semi-conducting materials	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure

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Electrical moisture absorption (extruded insulation only)	No failure

NOTE: Specimens rated for 2 kV test voltage shall be 2,000±50 Vrms between conductors. Fuses or overload relays shall each have a maximum rating of ½ A.

UNIT ORDERING LENGTHS

Type and size	Feet (nominal)
LS2KVTSGSG (all sizes)	1,000

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2017-009)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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